

INSTRUCTION 1: READ THE FOLLOWING QUESTIONS AND CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON THE SEPARATE ANSWER SHEET PROVIDED.

ANSWER SHEET PROVIDED.

1. A alves in the veins were discovered by:
A. Hieronymus Fabricius
B. William Harvey
C. Andreas Vesalius
D. Leonardo da Vinci
E. Hippocrate

2. Which one is not true about Anatomical Position?

- D
A. Body erect
B. Head, eyes and toes directed forward
C. Feet together
D. Arms at side with palms facing backward
E. None of the above

3. Flexion and extension movements are possible through:

- B
A. Sagittal axis
B. Horizontal axis
C. Vertical axis
D. Longitudinal axis
E. None of the above

4. Which one of the following movement is different from the others?

- A
A. Protrusion
B. Flexion
C. Extension
D. Abduction
E. Hyperextension

5. Which one of the following bone markings is not related to the humerus?

- D
A. Head
B. Facet
C. Condyle
D. Tuberosity

6. All of the following statements are true about sesamoid bone except:

- D
A. Prevent excessive wear and tear
B. Helps tendons to glide over bony surfaces
C. Alter the direction of pull of the tendons
D. Decrease mechanical advantage in the joints of long bones

E. None of the above

6. Which one is a correct order in structural organization of skeletal muscle from cell to whole muscle?

- A. Fascicle - muscle fiber - muscle
- ☒ B. Muscle fiber - fascicle - muscle
- C. Muscle - muscle fiber - fascicle
- D. Muscle fiber - muscle - fascicle
- E. None of the above

7. The plane that separates the body into unequal right and left sides is

- A. Transverse plane
- B. Midsagittal plane
- ☒ C. Parasagittal plane

- D. Frontal plane
- E. Coronal plane

8. The only bone that does not articulate with any other bone

- A. Sternum
- B. Nasal bone
- C. Vomer

- ☒ D. Hyoid bone
- E. None of the above

9. Which one of the following bone is not part of appendicular skeleton?

- A. Scapula
- B. Hip bone
- C. Clavicle

- D. Humerus
- ☒ E. Sacrum

10. A unique fibrous type of joint between a tooth & its socket

- A. Suture
- B. Syndesmoses
- ☒ C. Gomphoses

- D. Symphyses
- E. None of the above

11. Which one of the following is not a factor influencing synovial joint stability?

- A. The nature of the articular surfaces
- B. The number and positioning of the ligaments
- C. The tone and strength of the muscles acting upon the joint
- D. The type of joint
- ☒ E. None of the above

13. A type of synovial joint with a cylindrical shaped projection of one bone fits into a trough shaped surface of another bone with movement occurs within a single plane

A. Plane

B. Hinge

C. Condylloid

D. Saddle

* E. None of the above

14. Which one describes the condition in which an independent bones that fuse with other bones during development and remains as part of that bone?

A. Pressure epiphysis

B. Traction epiphysis

C. Apophysis

☒ D. Atavistic epiphysis

E. None of the above

15. The following are connective tissue layer around the skeletal muscle. Identify the correct order from internal to external

☒ A. Endomysium - perimysium - epimysium

B. Epimysium - perimysium - endomysium

C. Perimysium - endomysium - epimysium

D. Endomysium - epimysium - perimysium

E. None of the above

16. The ability of muscle tissue to return to original shape and length after contraction stretching is known as

A. Extensibility

B. Excitability

C. Contractility

☒ D. Elasticity

E. All of the above

17. The muscle fiber arrangement that we find around the openings of the body is:

A. Convergent

B. Parallel

☒ C. Sphincter

D. Pennate

E. All of the above

18. All of the following cranial nerves carry sensory, motor and parasympathetic except:

A. Facial nerve

☒ B. Oculomotor nerve

C. Glossopharyngeal nerve

D. Vagus nerve

E. None of the above

the internal surface of the body is:
Serous membrane
of the above

- D. Fixation, dehydration, infiltration, clearing & embedding
E. Clearing, dehydration, fixation, infiltration & embedding
25. Among the different lenses in a bright field microscope, which one functions in resolution of a tissue image?
A Objective lens
B Ocular lens
C. Condenser lens
D. a & b
E. b & c
26. To focus a blurred tissue image seen under a bright field microscope, you normally need to:
A. Change the objective lens to the appropriate one
B Move the stage carrying the tissue slide up or down
C. Move the objective lens up or down
D. Move the condenser lens up or down
E. Move the ocular lens up or down
27. The membrane of which organelle is continuous with the outer nuclear membrane?
A. Centrioles
B. Microtubule
C. Golgi apparatus
D/ Rough endoplasmic reticulum
E. Smooth endoplasmic reticulum
28. Vesicles produced from the rough endoplasmic reticulum will be transported to which organelle?
A. Smooth endoplasmic reticulum
B. Lysosome
C. Peroxisome
D/ Golgi apparatus
E. Microtubule
29. Which of the following is **not** a function of smooth endoplasmic reticulum?
A. Synthesis of steroid hormones
B. Detoxification
C. Store and release calcium ions in the striated muscle cells
D. All of the above
E/ None of the above

eight

19. A membrane which covers body cavities confined to the internal surface of the body is

- ☒ A. Mucous membrane
- B. Cutaneous membrane
- C. Meningeal membrane
- ☒ D. Serous membrane
- E. All of the above

20. Which one of the following statement is not correct concerning circulatory system?

- ☒ A. Capillaries are always located at the end of arterioles & beginning of venules
- B. Exchange of substances with extracellular fluid occurs at the capillary bed
- C. Arteries in the retina are true terminal arteries
- D. Pulmonary veins are atypical veins
- E. Lymphatic plexuses are absent in some tissues of the body

21. Which of the following show the appropriate respective section thicknesses for light electron microscope?

- ☒ A. 5 – 10 μm and 40 – 70 nm
- B. 40 – 70 nm and 5 – 10 μm
- C. 25 – 50 μm and 100 – 200 nm
- D. 100 – 200 nm and 25 – 50 μm
- E. e) 25 – 50 nm and 100 – 200 μm

22. Which of the following is a commonly used clearing agent in tissue section preparation for light microscopy?

- A. Ethanol
- B. Methanol
- C. Paraffin wax
- D. Formalin
- ☒ E. Xylene

23. In hematoxylin and eosin method of the general tissue section staining, hematoxylin

- A. Acts like an acidic dye
- ☒ B. Stains best the ribosomes
- ☒ C. Stains blue black
- D. a & b
- E. b & c

24. Which one of the following is the correct order of tissue block preparation autopsy tissue?

- A. Dehydration, fixation, clearing, infiltration & embedding
- B. Dehydration, clearing, infiltration, fixation, & embedding
- ☒ C. Fixation, dehydration, clearing, infiltration & embedding

C. Water molecules move from hypotonic to hypertonic solution

D. Sustained until isotonic solution is built

☒ E. Energy derives water form high to low water molecules

A 36. Which of the following is true about the fluid exchange between compartments

☒ A. The more the tissue colloid osmotic peruses is the less the fluid into the tissue

? ☒ B. The more the blood osmolarity is the more the fluid leave the tissue

C. Cell in the hypotonic solution losses water from intracellular compartment

D. The more the capillary permeability is the less the fluid get into the tissue

E. Osmotic pressure doesn't affect the fluid exchange between compartments

37. Which of the following is true about the organization of the human body

A. Only one type of tissue found in one organ

B. Cells in one type of tissue are different in function

C. Unrelated organs with function form one system

D. Atoms are functional components of the body

☒ E. Human body is merely the sum of the parts

38. Select the one which is true about the cells

A. Prokaryotic cells have organelles in different compartments

B. Membrane bounded organelles are found in prokaryotic cells

C. More than one chromosomes are found in one prokaryotic cells

☒ D. Organelles are found in the different compartments in eukaryotic cells

E. The outer and the inner membranes of eukaryotic cells are identical

39. Which of the following is true about the integral proteins in the cell membrane

A. Easily removed without destroying the cell membrane

☒ B. Serve as ion channels and receptors

C. Found in the inner membrane facing towards the cytoplasm

D. Not cross the whole cell membrane

E. Not penetrate the hydrophobic core of the membrane

40. Which of the following is not the function of the membrane proteins

A. Signal transduction

B. Communication

C. Recognition

24. Which of the following is wrong about the microtubules?

- A. Microtubule form 1) protofilaments arranged circumferentially
- B. Could increase in length, but not in width
- C. Transport vesicles within their luminal space from one region to another in the cell

25. Microtubules

- A. Nucleate themselves

26. Regarding the nucleolus:

- A. Facultative
- B. Site of mitochondrial synthesis
- C. Enzyme-bound region within the nucleus

27. Lamellar regions of chromatin contributed by different chromosomes

- B. Form visible under the light microscope

28. Organelles made by still of 5-10 discrete flattened and slightly curved bag-like chambers are:

- A. Mitochondria
- B. Smooth endoplasmic reticulum
- C. Peroxisomes
- D. Golgi bodies
- E. Microtubules

29. During cell cycle:

- A. Duplication of centrioles begins and complete within the G₁ phase
- B. The S phase is the longest and most variable phase

C. Daughter cells grow to the size of the parent cell during the G₁ phase

- D. Synthesis of DNA and proteins occur mainly during the S phase
- E. Replications of the chromosomes occur during G₁ phase

30. A mammalian body weight of 99kg could have _____ liter of plasma water

- A. 5400
- B. 10000
- C. 2800
- D. 380
- E. 2820

31. Which of the following does not characterize animals?

- A. Great concentration variation in their between compartments
- B. Water molecules move from high to low water molecules

D. Transportation

E/ None of the above

41. Why nucleus controls the cell activities

A. Found at the center of the cytoplasm

B. Being the largest organelle

C/ Contains genetic material

D. Having double membrane

E. Having nuclear envelop

42. Which of the following is true about transportation process across the cell

A/ Conformational change is the property of permeases

B. Energy brings conformational changes on the leaky ion channels

C. ATP is the direct energy in the secondary active transport

D. All carrier mediated transport is active

E. Binding site on the channels are specific

43. All of the following increase the diffusion rate of a substance across the membrane except one

A. The more the concentration gradient

B. The more the permeability of the membrane

C. The more the temperature of the substances

D/ The more the thickness of the membrane

E. The smaller the size of the substances

44. Which of the following is true about the Van't Hoff's law

A. The more the number of osmotically active particles is the higher the pressure

B/ The number of the osmotically active particle for non ionized chemical is 1

C. The osmotically active particle of urea is greater than the sodium chloride

D. Addition of calcium chloride into a solution reduces the osmotic pressure

E. The osmotic pressure of a solution with calcium chloride is less than the solution with sodium chloride with the same concentration

over a solution of $0.4\% \text{Na}_2\text{PO}_4$ with the following atomic weight: $\text{Na} = 23$, $\text{P} = 31$ and $\text{O} = 16$.

Then answer the following questions

45 Osmolarity of the solution

- A. 0.1175 osm/L
- C. 0.1567 osm/L
- D. 0.1650 osm/L

☒ B. 0.0850 osm/L

☒ E. 0.0485 osm/L

46 Osmotic pressure of the solution

- A. 1530.7 mmHg
- ☒ B. 1640.5 mmHg
- C. 5475.1 mmHg

D. 1563.5 mmHg

E. None of the above

47 Which of the following is not the property of the simple diffusion through passive channel?

- ☒ A. Transport maximum
- B. Dissipate potential
- C. Leaky ion channels

D. Hydrophilic open field protein

E. No energy needed

48 Which of the following is true about the active transport across the cell membrane?

- A. The source of energy for 2° active transport is indirectly ions
- ☒ B. Calcium influx into sarcoplasmic reticulum is 2° active transport
- C. $\text{Na}^+ - \text{K}^+$ Exchange in the cardiac cell is direct active transport
- D. Vesicular bulk transport is passive transportation process
- E. Pinocytosis is non energy mediated transport

49 Which of the following is not the function of the $\text{Na}^+ - \text{K}^+$ pumps

- A. Prevent intracellular edema
- ☒ B. Create sodium diffusion gradient
- C. Important for 2° active transport
- D. Creating membrane potential difference
- E. Avoid sodium gradient across the cell membrane

50 What happens to the rate of the transportation of a substance across the cell membrane in facilitated diffusion after transport maximum is achieved at a given time

☒ A. Constant

D. Slightly reduced

☒ B. Increased

E. No transport maximum in facilitated diffusion

C. Reduced

51. Which of the following is true about body fluid

- A. More water removed by kidney during exercise
- B. No water is lost from the body without regulation
- C/ Protein in the urine brings tissue edema
- D. The more the filtration coefficient is the less the filtration rate
- E. Capillary hydrostatic pressure gets high across the capillary

52. Which of the following is true about the composition of the intracellular and extra cell fluid compartments

- A. Intracellular osmolarity is greater than the extracellular
- ☒ B. More magnesium is outside the cell than the inside
- C. More proteins found inside the cell than the outside
- D. Potassium is the extracellular cation
- E. More water is found outside the cell than the inside

53. Which of the following electrolyte is found more in the blood vessels than in the tissue

- A/ Calcium
- B. Chloride
- C. Sulphate
- D. Bicarbonate ion
- E. Phosphate

54. The amino acid which can serve as proton donor and acceptor in

- A. Glycine
- B. Proline
- C. Histidine
- D. Isoleucine
- E. Tryptophan

55. A semi-essential amino acid that is required especially during pregnancy is

- A. Methionine
- B. Valine
- ☒ C. Arginine
- D. Leucine
- E. Tyrosine

56. The most common type of amino acid that frequently present in β -turns is

- A. lysine
- B. glycine
- C. valine
- D. isoleucine
- E. tyrosine

57. The electrostatic bond of side chains to form specific shape of a protein is described by

- A. Primary structure
- B. Secondary structure

☒ Primary structure

D. Quaternary structure

57. Which one is true regarding the folding of large polypeptides from polyribosomes complexes?

A. Folding starts in the ribosome

B. Its native folding is ATP independent

☒ C. It is a random and spontaneous process

D. It has high tendency for aggregation

☒ E. All of these

58. The role of molecular chaperones include the following except

A. They shield interactive surface of non-native polypeptide

B. Mostly expressed constitutively

☒ C. Recognize hydrophilic residues

D. They can reverse intramolecular misfolding

59. Facilitate assembly/disassembly of multiprotein complexes Chemical denaturing agents of proteins include

A. Ionizing radiation

D. Vigorous shaking

☒ B. Extreme pH

E. All of these

C. Heat

60. The level of protein structure disrupted during denaturation include the following except

☒ A. Primary structure

B. Secondary structure

C. Tertiary structure

D. Quaternary structure

E. Both C and D

☒ Organic compounds usually affect the protein structure by

☒ a. Affecting hydrogen bonding

☒ b. Disrupting hydrophobic interactions

c. Forming ionic bonding

d. Protonating key amino acid residues

e. Breaking peptide bonds

61. Which one is not true about denaturation of DNA?

☒ A. Covalent bonds will break

B. Physical changes because of high temperature

C. Hydrogen bonding is disrupted

~~B~~ A-T base pair more resistant than G-C base pair

~~E~~ Ionic interactions dissociate

64. Which one is not correctly matched?

A. Caudal/inferior

B. Ventral/anterior

C. Dorsal/posterior

D. Cranial/superior

~~E~~ None of the above

65. Wrong about thalidomide?

~~A~~ Amelia is a teratologic effect of thalidomide

B. Influenced for the study of teratology that with causes birth defects

~~C~~ Used as antineoplastic to pregnant women, resulting no birth defects

D. Affects the bones of the limbs

E. None of the above

66. Which one is not part of a uterine tube?

A. Isthmus

B. Infundibulum

C. Ampulla

~~D~~ Cervix

E. Intrauterine part

67. Which of the following is not found in the pelvis?

~~A~~ Testes

B. Prostate gland

C. Seminal vesicles

~~D~~ Ductus deferens

E. Ejaculatory duct

68. Which one is not found in the pelvis?

A. Cervix

B. Vagina

C. ovary

D. Uterus

~~E~~ None of the above

69. Wrong about primordial germ cells (PGCs)?

~~A~~ PGCs arise from the epiblast of the embryo

B. During their migration to the gonads they increase their number with mitotic division

C. Enter into meiotic division during the fourth – fifth weeks of development

~~D~~ Formed during the first week of development

E. Teratomas are tumors from abnormal development of PGCs

70. Wrong match about the stages of mitosis?

- A/ Metaphase—double structured chromosomes become visible
- B. Telophase—chromosomes uncoil and elongate
- C. Anaphase—Centromeres divide
- D. Metaphase—chromosomes migrate to opposite poles
- E. Prophase—Condensation of chromosomes

72. Wrong about the process of cross over?

- A. A chiasma is formed at the point of interchange of chromosomal segments of homologous chromosomes
- B/ It occurs at meiotic II
- C. Results genetic variability
- D. Random distribution of homologous chromosomes to the daughter cells
- E. None of the above

73. Wrong statement regarding birth defects?

- A/ About 50% of conceptions end in spontaneous abortion
- B. Non-disjunction of chromosomes can occur at meiosis I and II
- C. Cleft lip and palate are seen in trisomy 18
- D. Klinefelter syndrome contains extra chromosome number
- E. None of the above

Instruction II: Matching Questions. Match items from column "A" to items in column "B"

Column "A"

Column "B"

- 1. Transverse movement of phospholipids
- 2. Lateral movement of phospholipids
- 3. Preferentially component of inner leaflet of membrane
- 4. Sphingolipid component of myelin sheath
- 5. Modulator membrane fluidity

- A. Cholesterol
- B. Phosphatidylserine
- C. Phosphatidylcholine
- D. Sphingomyelin
- E. Fast movement
- F. Slow movement

Name: MAMA ABDUSAMAD

ID NO. 158/9649/107

I. Multiple choice

II. Match

1	A	25	CE	45	D	67	DA	1	F
2	D	26	E	46	B	68	E	2	E
3	B	27	A	47	A	69	D	3	B
4	A	28	B	48	BC	70	AD	4	D
5	D	29	D	49	BE	71	~	5	A
6	D	30	D	50	A	72	B		
7	B	31	E	51	E	73	AB		
8	E	32	EC	52	BC				
9	D	33	D	53	A				
10	E	34	D	54	E				
11	E	35	E	55	E				
12	E	36	BD	56	B				63
13	B	37	E	57	E				77
14	D	38	AC	58	ED				
15	A	39	ED	59	E				
16	D	40	D	60	B				
17	E	41	B	61	A				
18	B	42	E	62	AB				
19	AD	43	E	63	A				
20	A	44	A	64	E				
21	A		D	65	E				
22	E		B	66	D				

-1

-5

-5

-3

- A. Objective lens
B. Ocular lens

- C. Condenser lens
D. a & b
E. b & c

72. To focus a blurred tissue image seen under a bright field microscope, you normally need to:

- A. Change the objective lens to the appropriate one
B. Move the stage carrying the tissue slide up or down
C. Move the objective lens up or down
D. Move the condenser lens up or down
E. Move the ocular lens up or down

The question is the
as

Instruction II: Matching Questions: Match items from column "B" to items in column "A"

Column "A"

1. Transverse movement of phospholipids
2. Lateral movement of phospholipids
3. Preferentially component of inner leaflet of membrane
4. Sphingolipid component of myelin sheath
5. Modulator membrane fluidity

Column "B"

- A. Cholesterol
- B. Phosphatidylserine
- C. Phosphatidylcholine
- D. Sphingomyelin
- E. Fast movement
- F. Slow movement

Instruction III. Fill in the blank questions

1. Mannose and galactose are formed by epimerization of glucose.
2. glucuronic is a uronic acid formed by oxidation of C-6 of glucose into carboxyl group.
3. Arachidonic acid can be formed in our body from an essential fatty acid known as linoleic acid.
4. -Galactosamine is an example of amino sugar.